

The S1↔S2 Full-Attribute Swap

Does swapping all attributes of S1 and S2 at tStart work as a useful 50% manipulation?

1. Reference configuration (UNCUED trial, no swap)

Sub-field	Field	Depth + Color	Motion before tStart	Motion after tStart	Role
S0	A	Near+Red	aRot (coherent)	TRANSLATE (Linear)	Coherent translator
S1	A	Near+Red	NonCoherent	NonCoherent	Noise in Field A
S2	B	Far+Green	bRot (coherent)	bRot (coherent)	Coherent rotator
S3	B	Far+Green	NonCoherent	NonCoherent	Noise in Field B

2. Proposed S1↔S2 full attribute swap at tStart

At tStart, S1 takes all of S2's attributes (depth, color, motion type, direction) and vice versa.

Sub-field	Before tStart	→	After tStart	What changed
S0	Near+Red aRot → translate →		Near+Red TRANSLATE	Motion only (normal onset)
S1	Near+Red NonCoherent noise→		Far+Green bRot coherent	Depth, color, coherence all change
S2	Far+Green bRot coherent →		Near+Red NonCoherent noise	Depth, color, coherence all change
S3	Far+Green NonCoherent noise→		Far+Green NonCoherent noise	Nothing changed

3. Validity check — single color per depth plane?

- ✓ Near plane after swap: S0 (Red) + S2 (Red) = Near is Red throughout
 - ✓ Far plane after swap: S1 (Green) + S3 (Green) = Far is Green throughout
- Single color per plane maintained.

4. The problem — post-onset appearance is identical to no-swap

Compare what the observer sees after tStart:

Condition	Near plane (Red)	Far plane (Green)
No swap	S0 translates + S1 noise	S2 coherent rotates + S3 noise
S1↔S2 swap	S0 translates + S2 noise	S1 coherent rotates + S3 noise

Dots within a plane are physically indistinguishable.
The observer cannot tell whether the near-noise is S1 or S2,
or whether the coherent rotator in Far is S1 or S2.

5. Conclusion

- ✓ **Valid:** S1↔S2 swap maintains single color per depth plane — it does not violate the clean-plane rule.
- ✗ **Useless:** The post-onset stimulus is perceptually identical to no-swap. The observer sees the same functional structure: Near = translator + noise, Far = coherent rotator + noise. Only the anonymous labels on the dots change, which the observer has no access to.
- **Why:** S1 and S2 start in opposite planes. Swapping their attributes just moves one anonymous noise cloud to Far and promotes it to coherent, while demoting the existing coherent rotator to noise in Near. The result is structurally and perceptually equivalent to having done nothing.
- **Bottom line:** Any 50% swap that keeps planes clean either (a) produces a perceptually invisible rearrangement like this, or (b) changes both depth and color simultaneously — preventing independent manipulation of either. This is why 100% whole-field swaps are the minimum workable unit.